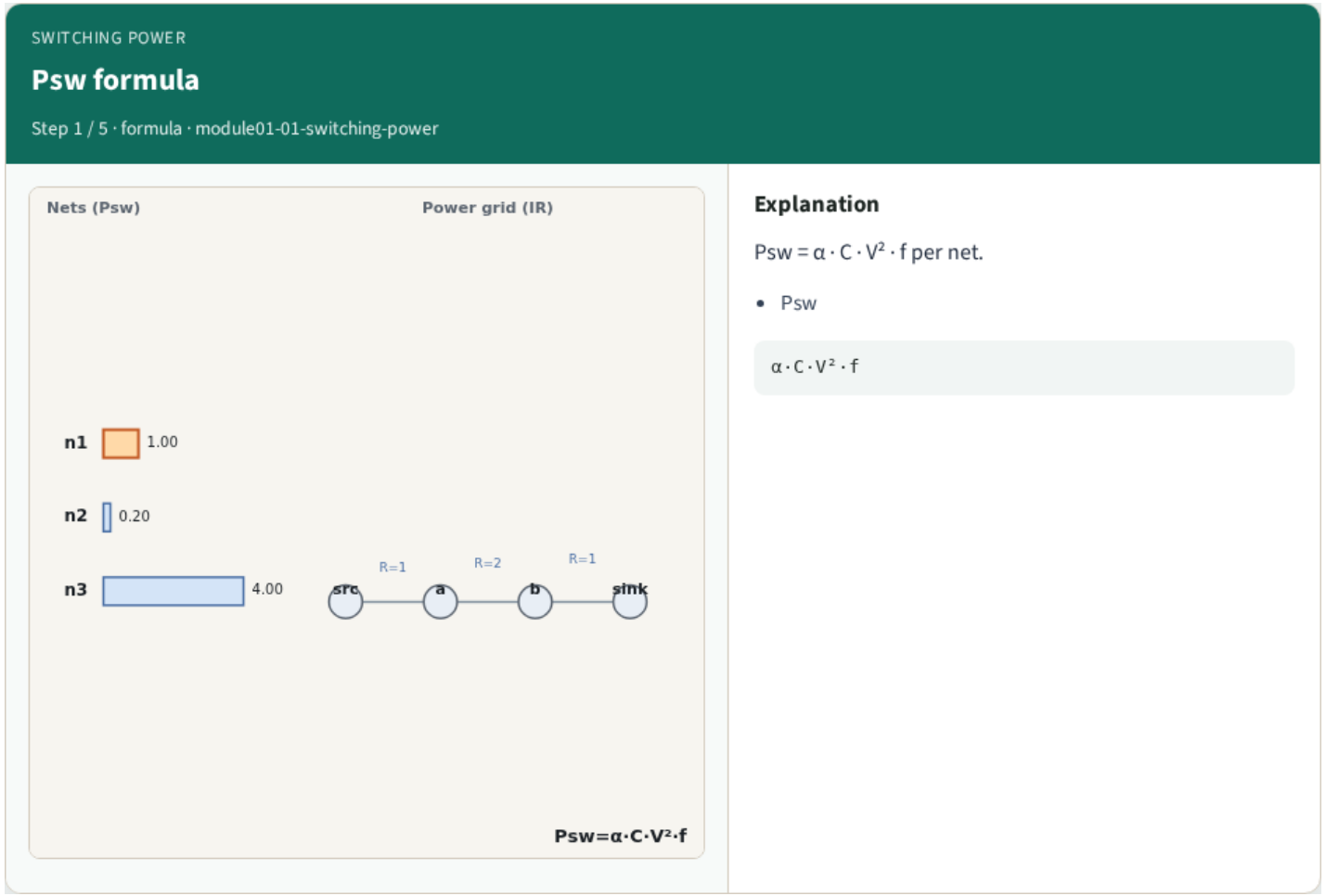
Why switching

Dynamic power scales with activity, cap, voltage squared, and frequency

The idea

- GOLDEN: psw_n1=1.0

Psw formula



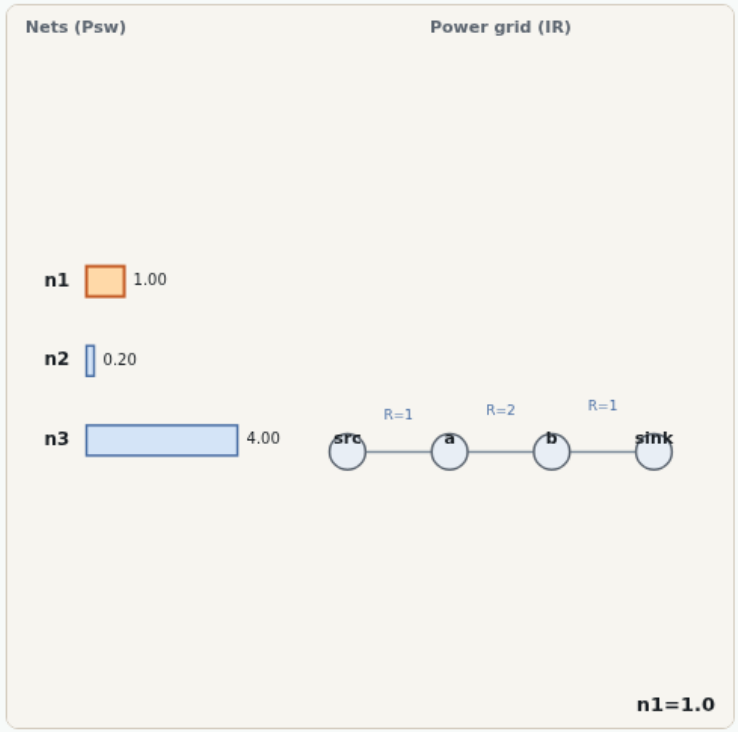
Psw formula

Net n1

SWITCHING POWER

Net n1

Step 2 / 5 · n1 · module01-01-switching-power



Explanation

n1: α=0.5, C=2, V=1, f=1 → Psw=1.0.

- n1

```
1.0
```

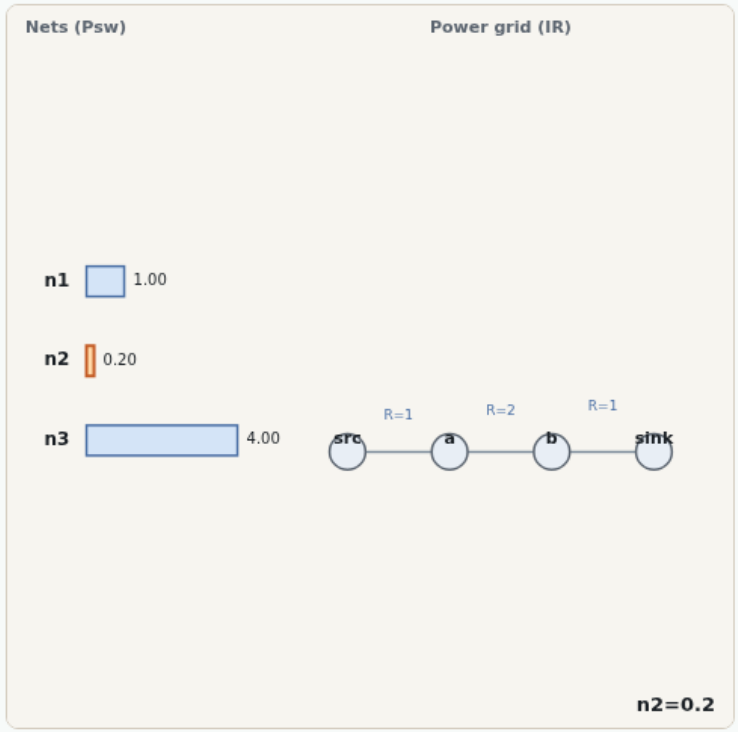
Net n1

Net n2

SWITCHING POWER

Net n2

Step 3 / 5 · n2 · module01-01-switching-power



Explanation

n2: α=0.2, C=1 → Psw=0.2.

- n2

```
0.2
```

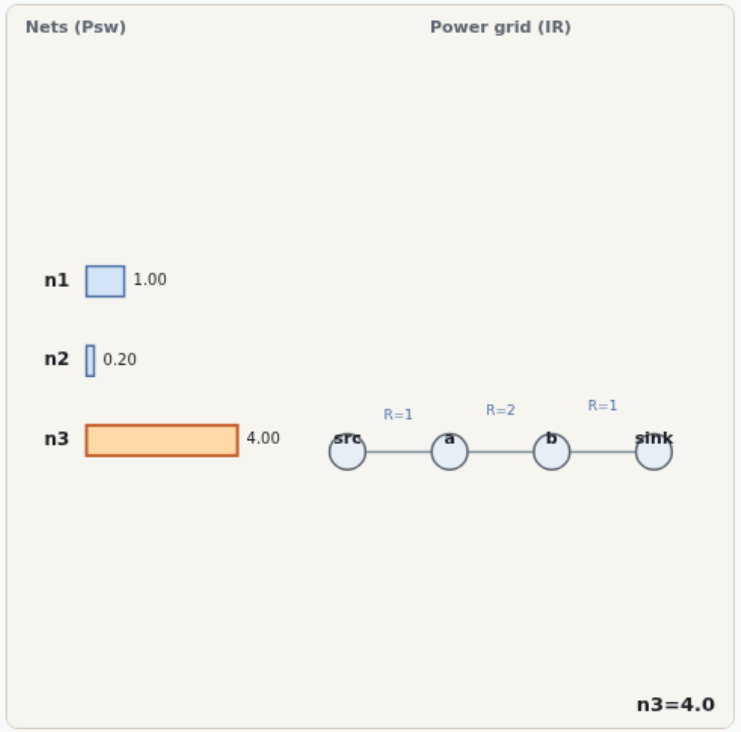
Net n2

Net n3

SWITCHING POWER

Net n3

Step 4 / 5 · n3 · module01-01-switching-power



Explanation

n3: α=1.0, C=4 → Psw=4.0.

- n3

```
4.0
```

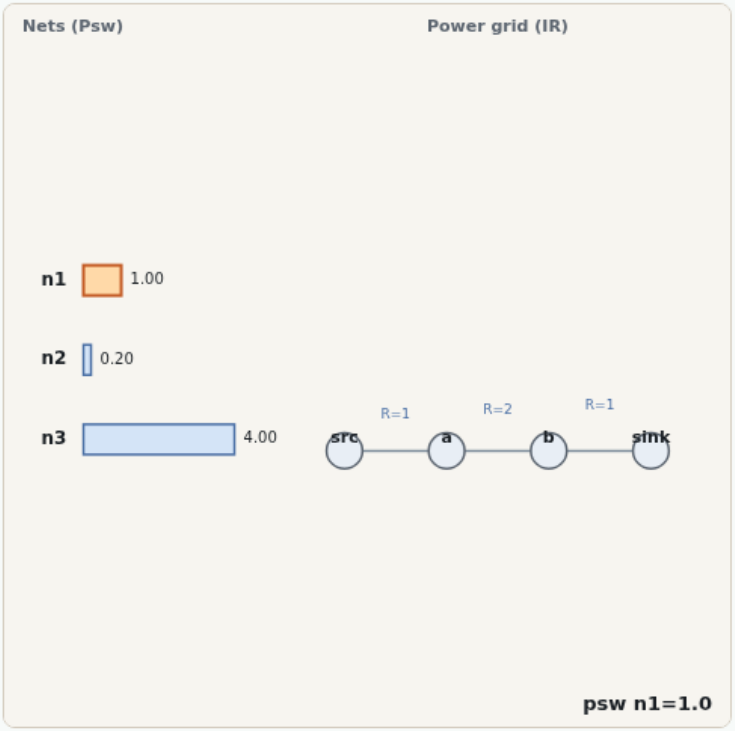
Net n3

Golden

SWITCHING POWER

Golden

Step 5 / 5 · golden · module01-01-switching-power



Explanation

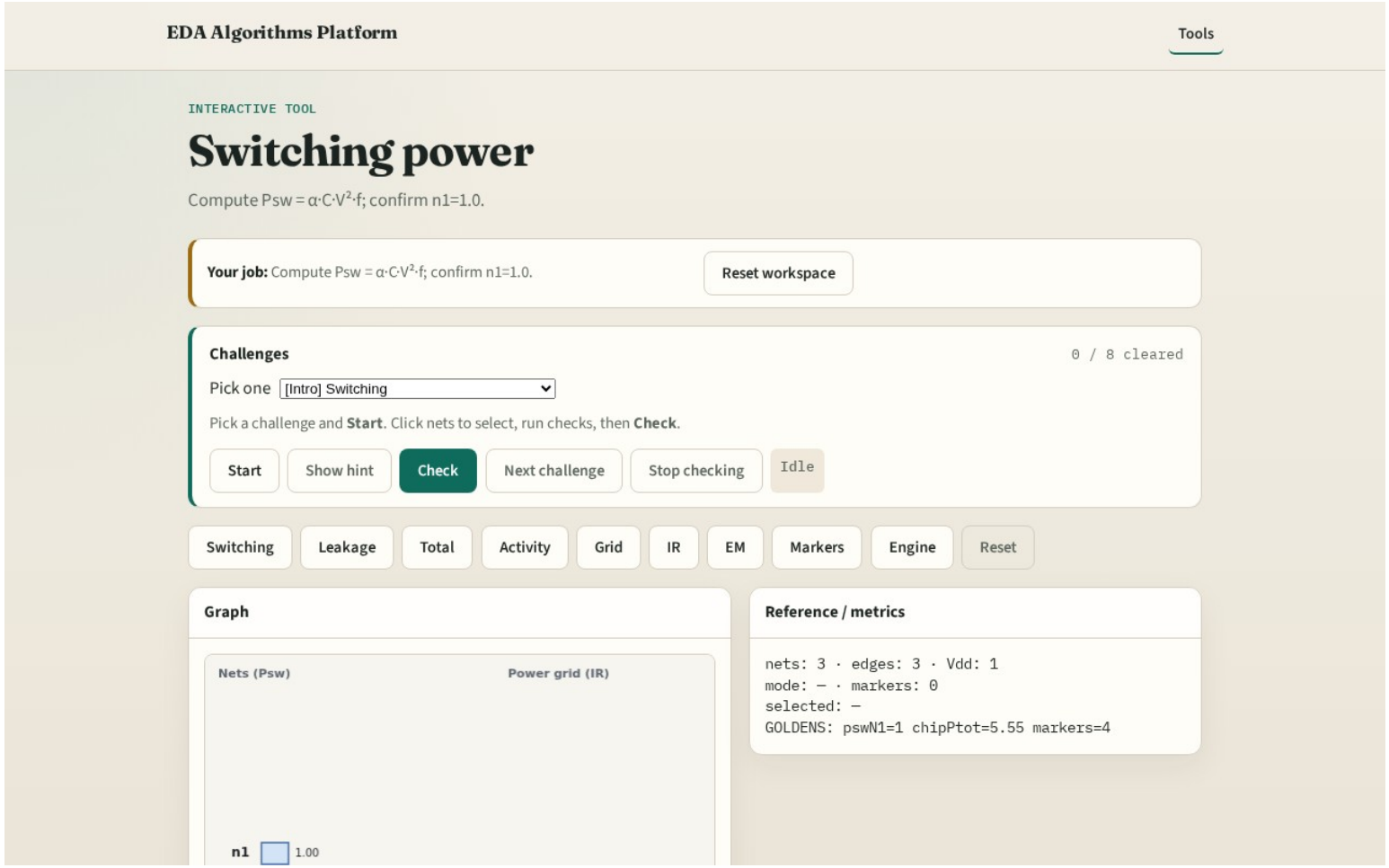
GOLDEN: psw_n1=1.0; chip_psw=5.2.

- GOLDEN

```
1.0/5.2
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call switching_power(); assert by_net['n1']==1.0

Pitfalls

- Forgetting V^2
- Using leakage instead of switching

Your turn

- Next: leakage power